Negative Points Generation Notes

Landslide early-warning project for Himachal Pradesh · M.Tech ICTD, IIT Delhi
Suraj Raj · Phase B (made 22–23 October 2026) · Notes written 2 November 2026

> **In one paragraph.** To teach a computer model when and where landslides happen, we need examples of landslides (*positives*) and examples of "no landslide" (*negatives*). Our first negatives were random points, mostly on high, icy, remote ground, so a model could tell them apart from landslides far too easily and learned nothing useful. We replaced them with 2,167 carefully chosen negatives of two kinds. **Same place, safe day** (969 points) teaches the model what *rain* does. **Similar place, storm day** (1,198 points) teaches it what the *ground* does. This document explains why, how, where the points are, and what the limits are.

1. Words used in these notes

Landslide record / positive	A landslide that really happened, with its location (latitude, longitude) and date. We have 981 of these from Himachal Pradesh (HP), Uttarakhand (UK) and Jammu & Kashmir (J&K). Label = 1.
Negative	A place and a date where *no landslide was recorded*. Label = 0.
Feature	A fact measured for each point, e.g. slope, rock type, distance to road, rain on the day, rain in the 7 days before. We use 42 features.
Model	A computer method that learns which feature patterns come with a landslide, and then gives any place and day a risk score from 0 to 1.
Google Earth Engine (GEE)	Google's free platform that holds satellite data and computes features for our points on Google's servers.
IMERG	NASA's satellite rainfall data (GPM IMERG). It gives rain for every half hour, worldwide. The "final" version (quality-checked) exists up to 30 Sep 2025 at the time of this work.
Buffer	A safety distance around something, e.g. "at least 2 km from any recorded landslide".
Median	The middle value when all values are sorted. Half are below, half above.
AUC	A score of how well a model separates landslides from negatives: 0.5 = coin toss, 1.0 = perfect.

2. Why do we need negative points at all?

A model learns by **comparing**. Think of a doctor learning which symptoms mean a disease: looking only at sick people is not enough, they must also see healthy people. Our landslide records are the "sick" cases. Negatives are the "healthy" cases.

The model can only learn the **differences between the two groups**. So the choice of negatives decides what the model learns:

- If negatives differ from landslides in silly, obvious ways (flat land, glaciers, far from everything), the model learns those silly differences.
- If negatives look like landslides in every way *except* "a landslide happened", the model is forced to learn the real causes: heavy rain and weak slopes.

> **The question a warning system must answer:** "Is *today* dangerous at *this* place?" That has two parts: **when** (which days are dangerous) and **where** (which slopes are dangerous). Good negatives must teach both.

3. What we had before: version 1 (random negatives)

How version 1 was made (October 2026, GEE script "Negative_points_generation", now archived)

1. Draw the outline of HP + Uttarakhand + J&K (from the FAO GAUL 2015 boundary map).
2. Throw 1,500 points at random inside that outline.
3. Remove any point within 2 km of a recorded landslide.
4. Keep 878 of the rest (one per landslide record at that time).
5. Give each point a date: a random landslide date, shifted randomly by up to ±30 days.

What was wrong with it (measured again in GEE on 29 Oct 2026)

	Old random negatives	Recorded landslides
Median elevation	**3,868 m**	1,228 m
Points above 4,000 m (snow, glaciers, bare rock)	**408 of 878**	10 of 969
Median distance to a motorable road	**4,773 m**	22 m
Points more than 2 km from a road	**535 of 878**	29 of 969
Points on gentle ground (slope under 10.9°)	145	about 10%
Points in the far north (above 35° N), where we have almost no records	209	almost none

- About two-thirds of the points fell inside the large J&K outline, much of it high mountains with no landslide records.
- The dates were copied from an early landslide list whose dates came from a faulty date-reading step (fixed later).
- Nobody checked whether a landslide happened near the point on its new date.

> **Why that is bad.** A model could separate the two groups with a lazy rule: "high, icy, flat or far from roads → no landslide". It would get a high score, but it would learn **nothing about rain**, which is exactly what a daily warning needs.

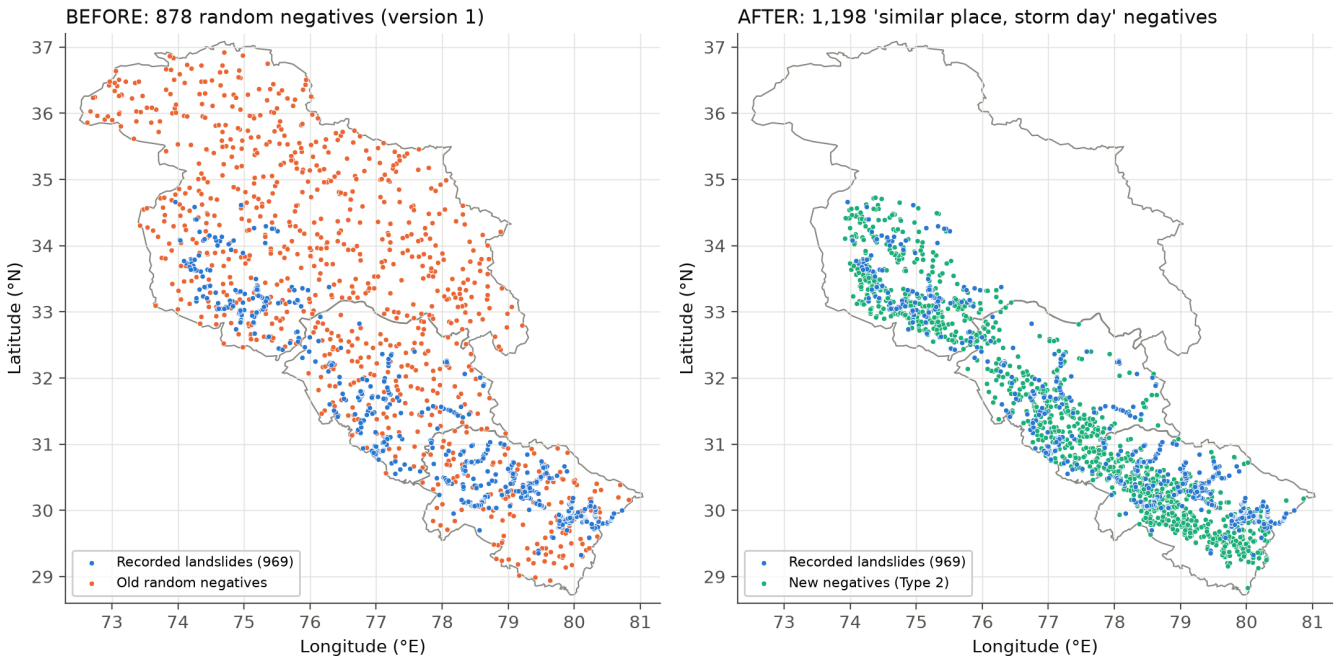


Figure 1. Left: the old random negatives (orange) are spread over the whole region, including the far north and high mountains where no landslides were recorded. Right: the new "similar place, storm day" negatives (green) sit in the same valleys as the recorded landslides (blue). The "same place, safe day" negatives are not drawn: they sit exactly on the blue dots. The grey outlines are the FAO GAUL 2015 boundaries used in the project.

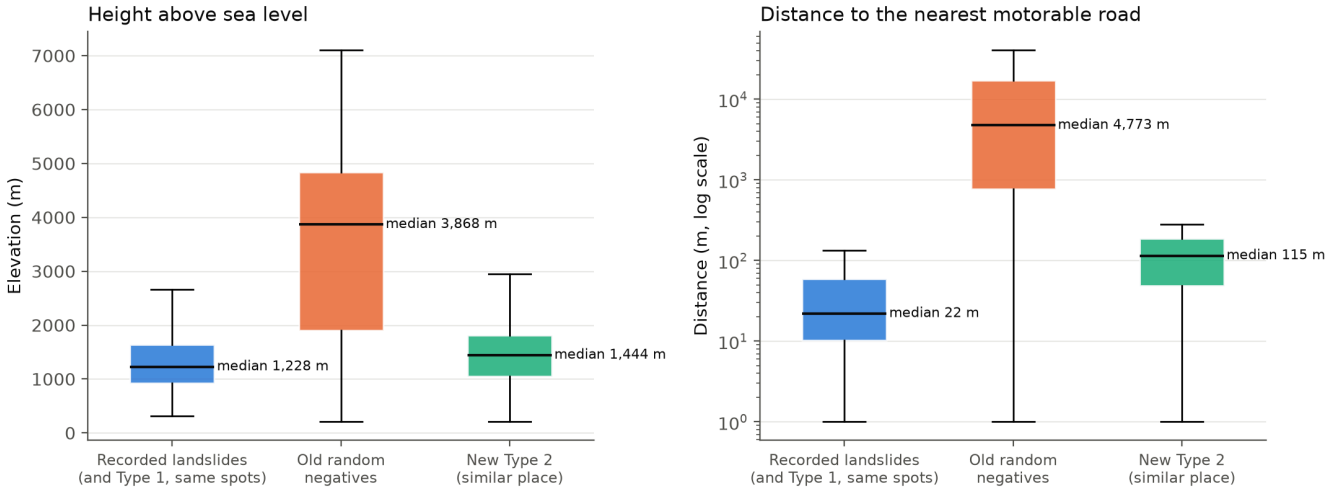
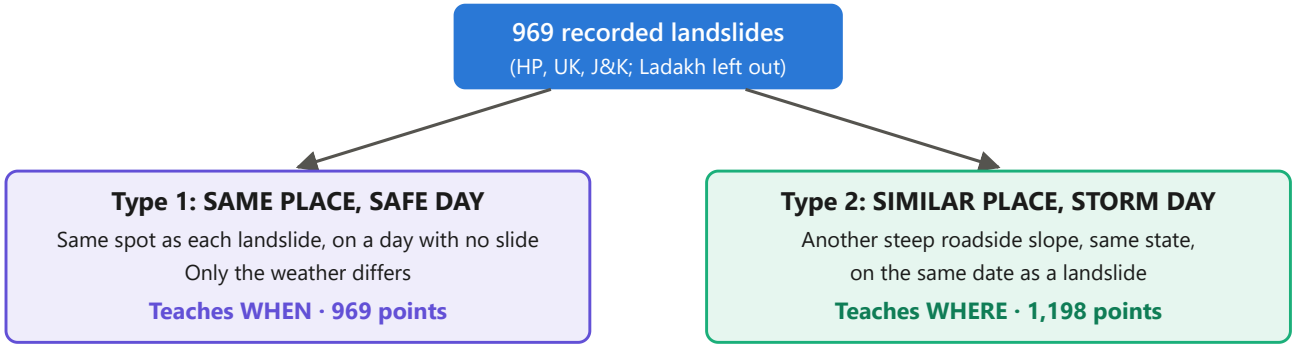


Figure 2. Boxes show the middle half of the values; the black line is the median; whiskers show the usual range. The old negatives were much higher and far from roads. The new Type 2 negatives look like the landslides. (Type 1 negatives have exactly the landslides' values, because they are at the same spots.)

4. The new idea: “hard” negatives

Instead of random points, make negatives that are **hard to tell apart** from landslides. We made two kinds, and each one teaches one part of the question.



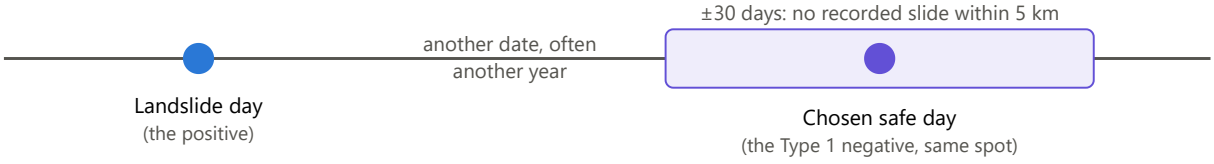
5. Type 1: Same place, safe day (teaches WHEN)

> **Idea.** Take the exact spot of each landslide and pick a different date when nothing happened there. The ground is identical, so the **only** difference the model can use is the weather.

The rules

Rule	Why
One negative per landslide, at the same coordinates (969)	Gives every landslide a “twin” on a safe day.
No recorded landslide within **5 km** and within **30 days before or after** the chosen date	Makes sure the chosen day really was quiet in that area, and allows for landslide dates in news reports being a few days off. (5 km and 30 days are our judgement, not values from a paper.)
The months follow the landslides’ month mix (88% June–September)	Most safe days are then **monsoon days**, often rainy. If safe days were mostly in winter, the model would only learn “monsoon = danger”. This way it must learn *how much* rain is dangerous.
Dates between 1 Jan 2005 and 30 Sep 2025	The period where quality-checked (“final”) IMERG rainfall exists.
Dates drawn from a pool of about 250 dates	GEE works date by date, so fewer different dates means much faster processing. In the end the 969 points use 244 different dates.
Terrain, rock, roads, land cover copied from the landslide’s row; only rain, soil moisture, snow and vegetation recomputed	The place is the same, so these values are the same. Copying them saved hours of computing.

Timeline at one landslide spot (example)



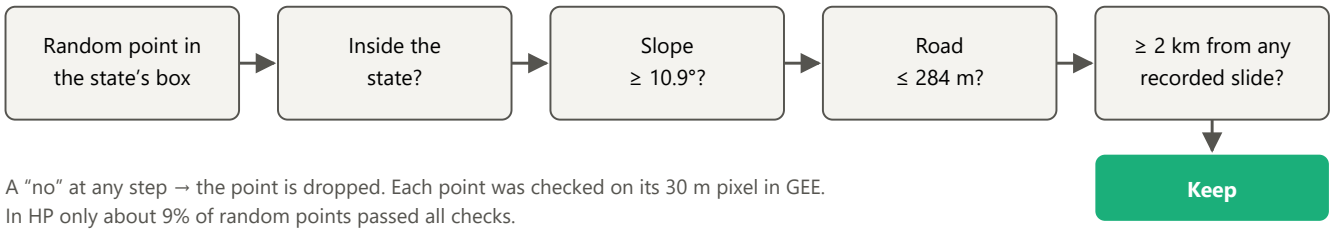
6. Type 2: Similar place, storm day (teaches WHERE)

> **Idea.** On the day of each real landslide, the storm was there. Pick other slopes nearby that look similar but did *not* slide. The weather is broadly the same, so the model must learn **which ground** fails: rock type, slope shape, closeness to streams and so on.

The rules

Rule	Why
Date = the date of a real landslide (the storm day)	Same storm conditions as the landslides.
Location: a random spot in the **same state** as a landslide on that date	Keeps the point in the region the storm affected. (A state is large, so the rain is not identical; see section 9.)
Slope at least **10.9°**	90% of our landslides are on slopes this steep or steeper. Flat land is an "easy" negative and teaches nothing.
Within **284 m** of a motorable road	90% of our landslides are this close to a road (because slides that hit roads are the ones that get reported). Negatives must be observed in the same way.
At least **2 km** from any recorded landslide (any date)	Reduces the chance of picking a slope that did slide but was never recorded. (2 km is our judgement.)
2 negatives per landslide, but at most **10 per date** (1,198 in total, 340 storm dates)	Without the cap, one huge storm (16 June 2013, Kedarnath disaster, 242 records) would fill the negatives with a single day.

How each candidate point was checked (in batches of 2,000 random points)



A "no" at any step → the point is dropped. Each point was checked on its 30 m pixel in GEE.
In HP only about 9% of random points passed all checks.

A practical note. We first asked GEE to sample whole states at 30 m resolution in one go, but it timed out (too big). So the points were generated in Python, checked by GEE in batches of 2,000, and the passing points were kept until every state had its quota. The rules stayed exactly the same.

7. Decisions that apply to both types

- **Ladakh is left out** of training: only 12 records, a cold-desert climate very different from HP, and not our target area.
- **Total: 2,167 negatives against 969 landslides** (about 2.2 to 1). Enough to learn from, without one group swamping the other, and small enough to compute in a few hours.
- **The same 42 features for every point**, computed with the same code as for the landslides. If the code differed, the model could learn the code difference instead of the real one.
- **Fixed random seed (20261022)**, so anyone running the scripts again gets exactly the same points.

8. How it was made, step by step

#	Step	Script / output
1	Generate Type 1 points (dates checked by the 5 km / ±30-day rule)	`phaseB_make_type1.py` → `negatives_type1_points.csv` (969 rows, 244 dates)
2	Generate Type 2 points (random points checked in GEE)	`phaseB_make_type2.py` → `negatives_type2_points.csv` (1,198 rows)
3	Compute features in GEE: 6 batch jobs (3 per type, split by date), run on Google's servers	`phaseB_extract.py` → GEE assets `negB_type1_a/b/c`, `negB_type2_a/b/c`
4	Join everything into one table; copy Type 1's fixed features from its landslide	`phaseB_assemble.py` → `training_table_v1.csv` (3,136 rows × 51 columns)
5	Checks (see below)	documentation §16, CHANGELOG

Checks done on the final table

- 3,136 rows: 969 landslides + 969 Type 1 + 1,198 Type 2. No duplicate IDs.
- Type 1: 969 of 969 have exactly the same fixed features as their landslide; 0 violations of the 5 km / ±30-day rule.
- Type 2: every point has slope ≥ 10.9°, road ≤ 284 m, nearest recorded slide ≥ 2.0 km, and at most 10 points per date.
- Missing values: only 2 cells in the whole table (snow on glacier cells, expected).

9. Where the new negatives are

	Recorded landslides	Type 1 (same place)	Type 2 (similar place)
Number	969	969	1,198
HP / Uttarakhand / J&K	195 / 548 / 226	195 / 548 / 226	324 / 487 / 387
Median elevation	1,228 m	1,228 m	1,444 m
Median slope	29°	29°	26°
Median distance to road	22 m	22 m	115 m
Points above 4,000 m	10	10	6
Most common years	2013, 2025, 2023	spread over 2005–2025	2025, 2015, 2023, 2013
Share of dates in June–September	88%	79%	82%

In HP, by district: landslides are mostly in Mandi (50), Solan (31), Shimla (25), Kinnaur (20) and Kullu (19). Type 2 negatives are mostly in Shimla (72), Mandi (69), Sirmaur (34), Kangra (32) and Chamba (26). So the new negatives sit in the **same middle-hill road valleys** as the landslides, not on glaciers or plains.

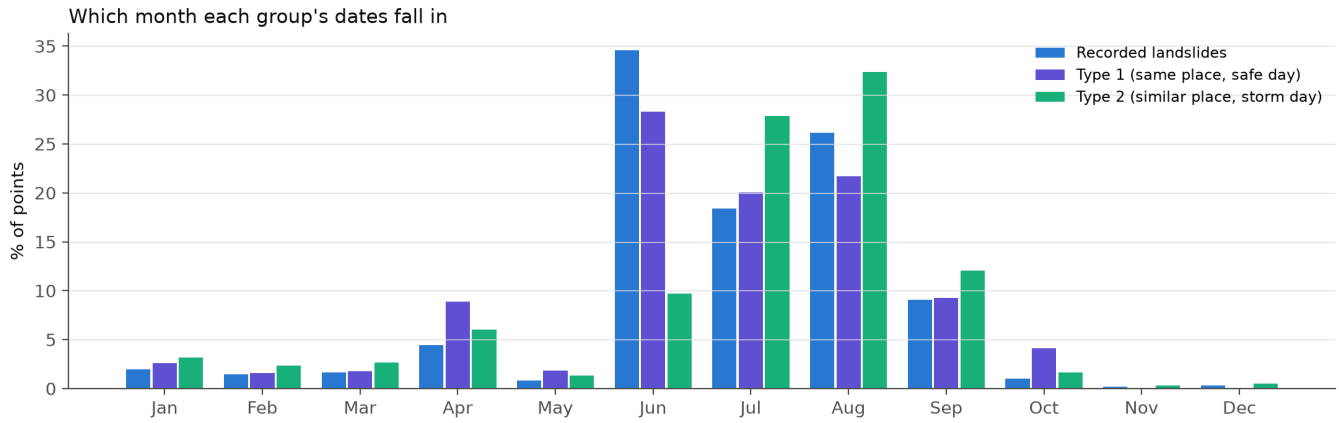


Figure 3. Both kinds of negatives are mostly in the monsoon months, like the landslides. Type 1 is a little lower in the monsoon (79%) because the "no slide within 5 km and 30 days" rule rejects more monsoon dates.

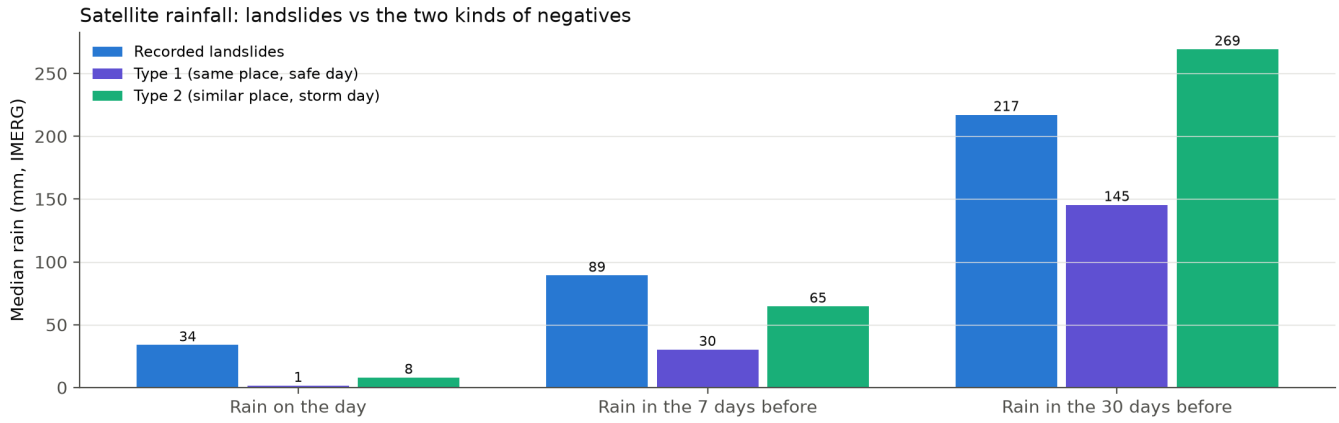


Figure 4. Median satellite rainfall. Type 1 (same place, safe day) is much drier than the landslide days: this is the "when" signal the model learns. Type 2 has similar rain over the past 30 days (same storm season) but less rain on the day: the landslides sit where the storm hit hardest.

10. What happened when the model was trained (Phase C)

- **Fair testing:** one whole year was hidden, the model trained on the other years and scored the hidden year, repeated for every year. This follows the warning by NASA's LHASA team that random splits make results look too good.
- **Overall:** the chosen model (Random Forest) scored an AUC of about 0.80.
- **"Weather-only" test using Type 1:** landslides vs same-place safe days, where only the weather differs. AUC about 0.75. This is the honest measure of how well the model reads rain, and it is only possible because of Type 1 negatives.
- **Road finding:** distance to road was among the most important features. Road features helped only against Type 2 negatives (Type 1 points have the same road distance as their landslide). This revealed the reporting bias described below, which is now being handled for the HP-wide map.

11. Before vs after

	Version 1 (random)	**Phase B (hard negatives)**
Number	878	2,167 (about 2.2 per landslide)
Location	random over the whole region; mostly high, remote, icy	the same spot (Type 1), or similar steep roadside slopes in the same state (Type 2)
Date	jittered from faulty dates; never checked	checked safe day (Type 1), or the real storm day (Type 2)
Features	slope and rain only	all 42, same code as for landslides
What a model can learn	"mountain top = safe" (useless for warnings)	how much rain is dangerous (when) and which slopes fail (where)
Can we measure rain skill separately?	No	Yes (the Type 1 test)
Repeatable	partly	fixed seed; scripts, task IDs and checks saved

12. Limitations (to state honestly)

1. **"No landslide recorded" is not "no landslide happened."** Some negatives may be landslides nobody reported (called *label noise*). The 2 km and 5 km buffers reduce this but cannot remove it.
2. **Everything is near roads.** Landslides get reported mostly where they hit roads, so our landslides and Type 2 negatives are all close to roads. A model trained on this can wrongly think "far from a road = safe". This is a known problem in landslide studies (Steger et al.). The planned fix for the HP-wide map (being tested in November 2026): set the road features to a typical roadside value everywhere, and mark remote areas as "lower confidence".
3. **Type 2 is in the same state, not the same storm cell.** Its rain on the day is lower than at the landslides (median 8 mm vs 34 mm), so some of the "where" signal is mixed with rain differences.
4. **Some thresholds are judgement values** (5 km, ±30 days, 2 km, 2 per landslide, 10 per date). The slope (10.9°) and road (284 m) thresholds come from our own data.
5. **Type 1's monsoon share** came out at 79% instead of the planned 88%.

13. Research basis

- **Steger et al. (2024)**, space–time landslide prediction, *Geoscience Frontiers* 15(3); and the ICG2022-388 abstract by Steger, Moreno, Crespi et al.: absence observations are taken both "at landslide locations and at non-landslide locations". This is our Type 1 + Type 2 design.
- **Zhu et al. (2024)**, *Geomatics, Natural Hazards and Risk* (Huize County, China): negatives chosen with buffer rules gave better models than purely random sampling.
- **Stanley et al. (2021)**, NASA LHASA version 2, *Frontiers in Earth Science* 9:640043: random space–time negatives; warns that random train/test splits overstate skill. We therefore test on hidden years.
- **Steger, Brenning, Bell & Glade (2017)**, *Landslides* 14:1767–1781; and EGU 2018 abstract "Incompleteness matters": landslide inventories are biased towards places people look (e.g. near infrastructure), which can mislead models.
- **Our own data:** the 88% monsoon share, the 10.9° slope threshold and the 284 m road threshold were measured from our 969 landslides.

14. Summary

> Our first negatives were random points, mostly high, icy and far from roads, so a model could separate them trivially and learned nothing about rain. We replaced them with two kinds of hard negatives. **Same place, safe day**: each landslide location on a checked landslide-free date, with months matched to the monsoon. It isolates the rainfall signal. **Similar place, storm day**: steep roadside slopes in the same state, on the same storm date, at least 2 km from any recorded slide. It isolates the terrain signal. That gives 2,167 negatives, about 2.2 per landslide. They let us test rain skill on its own (AUC about 0.75), and they revealed the road-reporting bias that the project now corrects. The main limitation is that some negatives may be landslides nobody recorded.